

Logan Lane

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EDUCATION

Virginia Tech

Ph.D. in Computer Science and Applications advised by Doug A. Bowman
Thesis: *Improving 3D Interaction Through Models of Task Completion Time*

Blacksburg, VA, USA

Aug. 2021–Aug. 2026 (Expected)

Virginia Tech

M.S. in Computer Science and Applications - *Earned en route to Ph.D.*

Blacksburg, VA, USA

Aug. 2021–May 2024

University of Virginia's College at Wise

B.S. in Management and Information Systems
Minors in Computer Science & Business Administration

Wise, VA, USA

May. 2018–May 2021

EXPERIENCE

Virginia Tech

Graduate Research Assistant

Blacksburg, VA, USA

May. 2022–Current

- Conducted research that developed and validated performance models to predict user interaction time for 3D User Interfaces through a grant from the Office of Naval Research
- Conducted additional VR/AR research spanning a range of topics including collaboration, training, and general 3D interaction techniques
- Developed prototype VR/AR software using C#, Unity, Unreal Engine, and Photon/Normcore Networking technologies for Meta Quest and Magic Leap Head Worn Displays
- Investigator on multiple HCI studies; recruited and ran nearly 200 participants

Prosimulation, LLC.

Founder/Developer

Blacksburg, VA, USA

March. 2023–October 2025

- Served as PI on a \$50,000 proof-of-concept grant and directed a five-developer team
- Achieved 90.1 mAP@50 detecting football players by fine-tuning RF-DETR on 2,000 annotated frames; 84.4% precision and 86.0% recall
- Built a VR analysis tool that ingests RF-DETR outputs to reconstruct and visualize player trajectories in 3D, synchronized to game film for film review and training in an immersive, 3D environment

NASA

Human Factors Engineer Intern

Huntsville, AL, USA

June. 2024–Aug. 2024

- Developed VR software to simulate lighting conditions on the south pole of the Moon for astronaut training simulation prior to NASA's Artemis mission launch
- Created interactions for Lunar sample collection, shuttle inspection, and photography in VR training simulation using Unreal Engine 5
- Received certificate of recognition for created interactions and mentoring other interns throughout the summer

PUBLICATIONS

Journal Papers

1. **Logan Lane**, Feiyu Lu, Shakiba Davari, Robert J. Teather, and Doug A. Bowman, "Revisiting Performance Models of Distal Pointing in Virtual Reality", *IEEE Transactions on Visualization and Computer Graphics*, 2025

- Alexander Giovannelli, Jerald Thomas Jr., **Logan Lane**, Francielly Rodrigues, and Doug A. Bowman, “Gestures vs. Emojis: Comparing Non-Verbal Reaction Visualizations for Immersive Collaboration”, *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2023

Conference Proceedings

- Logan Lane**, Ibrahim Tahmid, Feiyu Lu, and Doug Bowman, “Evaluating the Viability of Additive Models to Predict Task Completion Time for 3D Interactions in Augmented Reality”, *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 2026
- Logan Lane**, Jerald Thomas, Alexander Giovannelli, Ibrahim Tahmid, and Doug Bowman, “Exploring the Effects of Level of Control in the Initialization of Shared Whiteboarding Sessions in Collaborative Augmented Reality”, *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 2025
- Emilee Lane, **Logan Lane**, Diane Millar, Keith Gentry, and Michelle Lenhart, “Virtual Reality as an Intervention for Individuals Diagnosed with Autism Spectrum Disorder”, *American Speech-Language-Hearing Association (ASHA) Convention*, 2024
- Jerald Thomas, Sang Won Lee, Alexander Giovannelli, **Logan Lane**, and Doug Bowman, “A Communication-Focused Framework for Understanding Immersive Collaboration Experiences”, *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2023

Extended Abstracts

- Logan Lane**, Alexander Giovannelli, Ibrahim Tahmid, Francielly Rodrigues, Cory Ilo, Darren Hsu, Christos Lougiakis, Shakiba Davari, and Doug A. Bowman, “The Alchemist: A Gesture-Based 3D User Interface for Engaging Arithmetic Calculations”, *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 2024
- Alexander Giovannelli, Francielly Rodrigues, Shakiba Davari, Ibrahim A. Tahmid, **Logan Lane**, Cherelle Connor, Kylie Davidson, Gabriella N. Ramirez, Brendan David-John, and Doug A. Bowman, “CLUE HOG: An Immersive Competitive Lock-Unlock Experience using Hook On Go-Go Technique for Authentication in the Metaverse”, *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 2023

SKILLS

Programming Languages: C#, Python, Swift, SQL, Kotlin

Development Tools: Unity 3D, Unreal Engine, Photon PUN/Fusion, Normcore, Git, Xcode, Android Studio

ML, Data & APIs: PyTorch, vLLM, FastAPI, REST APIs, DuckDB, Docker, CUDA, SQLite

Platforms: Meta Quest, Magic Leap 2, Vision Pro, Linux, macOS

MENTORING

Undergraduate Researchers

- | | |
|---------------|----------------|
| 1. Ben Tatum | 2025 - Present |
| 2. Darren Hsu | 2023 - 2024 |

Other Mentoring Experience

- | | |
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| 1. Numerous undergraduate students | |
| Student Support Services at UVA Wise (Student Mentor) | 2019-2021 |
| 2. Numerous high school students | |
| Upward Bound at Southwest Virginia Community College (Instructor & Teaching Assistant) | 2017-2019 |

TEACHING

Graduate Teaching Assistant at Virginia Tech

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|---|-----------|
| Mobile App Development, Database Management Systems, Professionalism in Computing | 2021–2023 |
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SERVICE & PROFESSIONAL DEVELOPMENT

Reviewer

ISMAR 2024 (Conference Track), ISMAR 2025, CHI 2025, IEEE VR 2025, CHI 2026 (Poster Track)

Student Volunteer

ISMAR 2025, IEEE VR 2023 (Satellite Event)

SCHOLARSHIPS AND AWARDS

LAUNCH Proof of Concept Grant - \$50,000	2025
RADM Fred Lewis Scholarship at I/ITSEC 2024 - \$10,000	2024
UVA Wise Unsung Hero Award	2021
TICE Total Achievement Award	2020
Dr. Anne Crum Student Leadership Award	2015